



## Strategic Use of Color in Visuals Transcript

Hello everyone, welcome to BITS Pilani Digital. We will continue our learning journey for the course Data Visualisation and Storytelling. Today we will continue our effective visual design topic and look at a specific topic of strategic use of colours in visuals.

I cannot stress the importance how visual colours have to be used effectively to bring more clarity and aesthetics to our visualisation. In this regard, we will be using colour as a pre-attentive attribute to bring the focus and attention on to the visual information being presented in the graphic. Now let us take step by step about how we can use colours as a strategic element in our visuals and get the focus of our audience.

Take a look at the graphics that are on the slide. One is the palette which is heavily coloured trying to showcase the country level sales ranks of top five drugs. However, the information contained in the table is being marred by the colours that are used.

It is not very clear though it shows some information it is not readily perceivable. Most importantly, we can say it is causing clutter or higher cognitive load to process the information. As opposed to this if the same palette is used different form by colour saturation as showcased at the right end, the information is readily followed.

You can see one can read into the graphic with much less effort and draw some conclusions. This is one of the cases where colour can act against the effectiveness of communication and it has to be used strategically though it looks less colourful it is actually more pleasing and driving the information to the audience. One should use the colour very effectively and with a specific purpose in mind.

In this regard colour should be used sparingly and it should be used with an intention unintentional use of colour just to highlight just to create an impact should be avoided at all costs. Colour can be a very good friend but also can destroy message if it is used in excess or in bad combinations. So follow the following four principles while using colour be consistent whatever colour you are using for a particular element in one graphic should be carried on to other graphics also so that there is some visual continuity.

Let us say you are colouring India in blue and you can use the same blue colour for highlighting text of India also there is some visual relationship between these two that the human mind can relate to and that can be a very good aesthetic feature to the visualisation. Similar consider the problems of the colour challenged people who can have difficulties in perceiving different shades of colours. Also understand the emotional tone for example you have a sad reporting like number of deaths in your visual.

There is no point in showing that particular visualisation in bright and joyful colours it should reflect the tone of the visual. Finally use the brand colours very thoughtfully these days branding is a very potent exercise to reach out to the audience and there

are specific colours and fonts that are built by companies to reach out to their audience or reach out to their customers. When we are using the brand present when we are doing brand presentations or making presentations within a company ecosystem usage of specific colours is mandated in such instances one has to be very careful about how to assimilate those brand colours without destroying the meaning that we want to convey in our visual.

In cases where we feel the brand colours need to be used we should use them otherwise we should take a call. We will look at these individual aspects closely with the help of examples. Now we look at consistency of colours generally when we see the colour palette we have a natural ordering of colours or it is intuitively some colours are placed with certain signs or categories.

So for example when we look at the colours here blue red green yellow this is the sequence in which you see the colours in the colour palette in an Excel or somewhere. So naturally ordering north south east west which is also an order in its own in this form gives a consistent use of colour. Now again if you use the same colouring scheme in a different format in the same visualisation and changing the colours is a bad practise.

It gives confusion it creates confusion for the audience. This should be avoided. As I mentioned consistent use of colours whenever feasible wherever feasible should be adopted.

So one should avoid novelty for its own sake you should not use colour or creating a colour schema only to highlight certain aspects which need not be highlighted otherwise. So whenever we are using colour it should signal a meaningful message and we have to use colour to get attention only when it is needed just to put too many attentive points on the graphic can also lead to confusion. Remember Gestalt focal point principle works when there is light background or when there is similar background and you want to highlight a couple of points.

If there are too many attentive elements on the graphic the message is lost. Similarly preserve the colour meaning I was just saying this earlier if you have assigned a category a particular colour try to continue it across the graphic or the story for example in a presentation back you have started one particular category of colourings for different sales regions try to carry them towards the end of the presentation so that the recall is very strong and it can be associated with that particular colour and this helps in building a intuitive storyline and people can quickly relate to the colour and to the region that you are trying to put it. Finally you can use colour to reinforce structure or change this has to be used very strategically when you are trying to show red as growth red as degrowth green as positive growth it can be done with a particular intent but not arbitrarily you have to give lot of thought before using any specific colour in your graphics and always try to be consistent with the use of colours.

Next we will talk about accessibility this is a very important paradigm how often lost in the urgency to build visuals remember one of the key tenets of designing good visuals is getting broader acceptance and acceptance is built on accessibility you should make your visual accessible to a broader range of citizens this happens when you are sensitive to their requirements again this follows the philosophy that your visualisation should be audience centric but not data centric you should understand the perceivers notion of your visual you should make it broadly accessible in this context you should also look out for the colour challenged people who might not be able to pursue different shades of colours for them you should avoid certain principles for them you should follow certain principles and avoid certain colour combinations so that the graphics are accessible to a broader range of people now look at the graphics on the slide the first graphic is more challenging for colour challenged people compared to the other graphic where it uses colour saturation to bring out the same meaning the meaning has not lost but the effective visualisation ensures it is accessible to a broader range of audience similarly in the next graphic you have used red and green with a meaning but it would be better to use blue orange combinations with visual signs of 10% and minus 8% to drive home the meaning for a broader range of people. So what are the general rules avoid green and red combinations these are most problematic for colour challenged people and should not be used in visual comparisons per se you can also use supplementary visual cues if you cannot get away with the colours you can use plus and minus symbols or additional cues to drive home the meaning bold text brightness size can be used to help people understand and access the graphic similarly alternate colour schemes can be used for blue and red and green like blue for positive orange for negative this can be also considered if it permits you should be using colours always that are friendly finally there is no need to show both the extremes if you show one positive you can leave out the other negative so this kind of considerations will help people make the graphic more accessible and broad reaching again whenever it is necessary you can use certain colours but otherwise always keep the universality of the graphic in mind while designing the visuals just to mention there are certain softwares which will help you test whether your visual is colour friendly or not I mean whether it is friendly for a larger rate of audience or not. Now coming to one of the most important topics of use of colour we mindlessly apply colours to our visuals thinking that they build attention into the visual which is true also but they have to reflect the tone that we are trying to convey in our message in some of our settings the messages are very objective and need not carry a meaning for example you are discussing about sales growth you can use green for growth or blue for positive growth and orange for negative growth and make it appealing but even if you do with other colours it should not matter whereas if you are reporting let us say increase in crime rate or increase in share market where there might be some emotion it is always better to use colours that reflect such tone and motion such tone and context it has to be kept in mind that usage of wrong colours at times might irritate the audience and invite



frictions this can be avoided if you can use appropriate saturations and colour combinations which are right for the context of the visual.

Let me give you an example let us say you are talking about passion energy excitement it is better to use some shades of orange and red similarly you are hoping growth positivity you can use green integrity blue these are well established colour palettes that are given in the visualisation domain that can be readily picked up for reflecting the mood and wrong colours can definitely irritate audience but also cause frictions that might lose the message that you want to convey so think about the colour palette that you are using whether it is suitable for the tone that you want to convey so assure yourself this is the right colour and then only use them

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